

OLIST E-COMMERCE ANALYTICS DASHBOARD

Driving Business Insights through Data Visualization

PRESENTER

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DATASET

Brazilian Olist E-Commerce

TRANSFORMING RAW DATA INTO ACTIONABLE BUSINESS INTELLIGENCE

Primary Objective

To analyze real-world e-commerce transactions and deliver interactive insights that support data-driven decision-making for business stakeholders.

Analytical Approach

Using an end-to-end pipeline that transforms fragmented operational data into a unified, visual narrative of business performance.



Data Cleaning & Preprocessing



Exploratory Data Analysis (EDA)



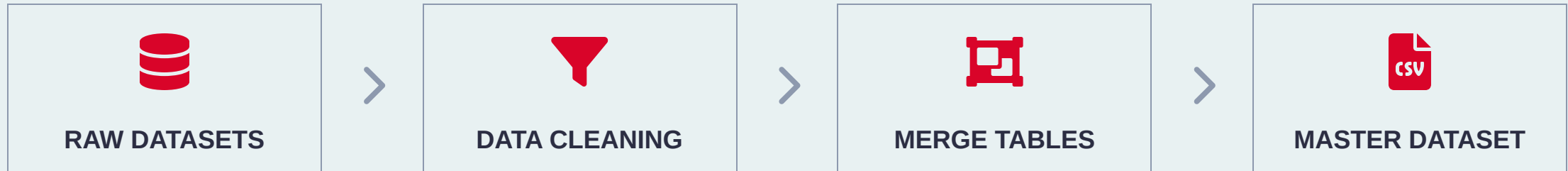
Business Metric Identification



Interactive Streamlit Dashboard

LEVERAGING THE BRAZILIAN OLIST DATASET FOR LARGE-SCALE ANALYSIS

The Olist dataset provides a comprehensive view of the Brazilian e-commerce landscape, containing over 100,000 anonymized orders from 2016 to 2018. It integrates data from multiple marketplaces, including orders, customers, sellers, products, and reviews.



RIGOROUS DATA PREPARATION ENSURES ANALYTICAL ACCURACY

DATA CLEANING

-  **Duplicate Removal:** Eliminated redundant order entries to prevent revenue overestimation.
-  **Missing Values:** Imputed or removed null values in critical logistics and review columns.
-  **Type Conversion:** Standardized 8+ timestamp columns into datetime objects for time-series analysis.

FEATURE ENGINEERING

-  **Delivery Performance:** Calculated actual delivery time vs. estimated delivery windows.
-  **Temporal Features:** Extracted Purchase Hour and Weekday to identify peak demand cycles.
-  **Logistics Flags:** Created a binary 'Late Delivery' indicator to correlate with customer sentiment.

ANSWERING CRITICAL BUSINESS QUESTIONS THROUGH VISUALIZATION



Revenue Performance

Which sellers and product categories generate the highest total revenue?



Customer Satisfaction

What is the overall distribution of customer review scores?



Temporal Trends

Which weekdays and purchase hours experience the highest order volume?



Logistics Efficiency

How long do deliveries typically take, and what is the frequency of late orders?



Impact Analysis

Does late delivery significantly correlate with lower customer review scores?



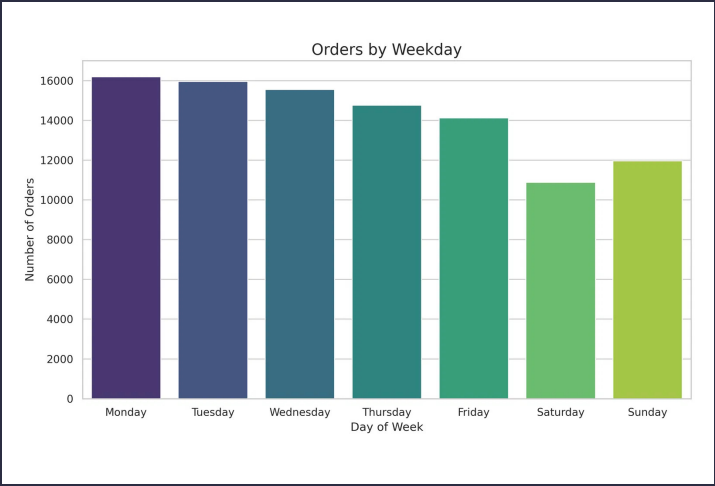
Geographic Insights

Which Brazilian states contribute most to the platform's sales and growth?

EVERY QUESTION ADDRESSED USING INTERACTIVE PLOTLY VISUALIZATIONS

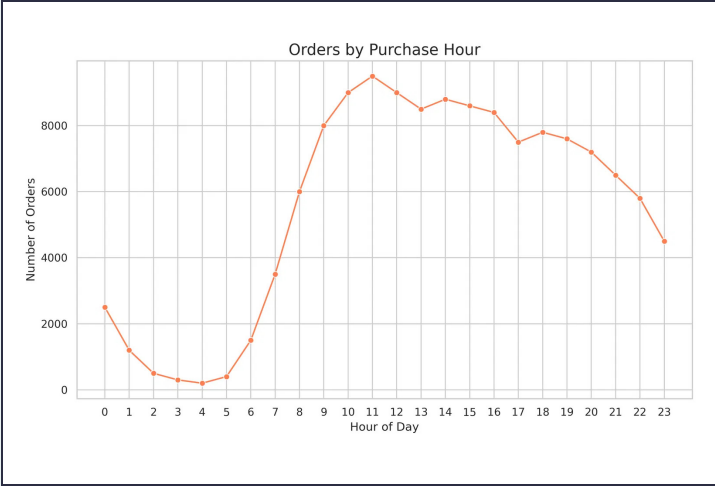
MONDAY PEAKS AND AFTERNOON SURGES DEFINE ORDER PATTERNS

ORDERS BY WEEKDAY



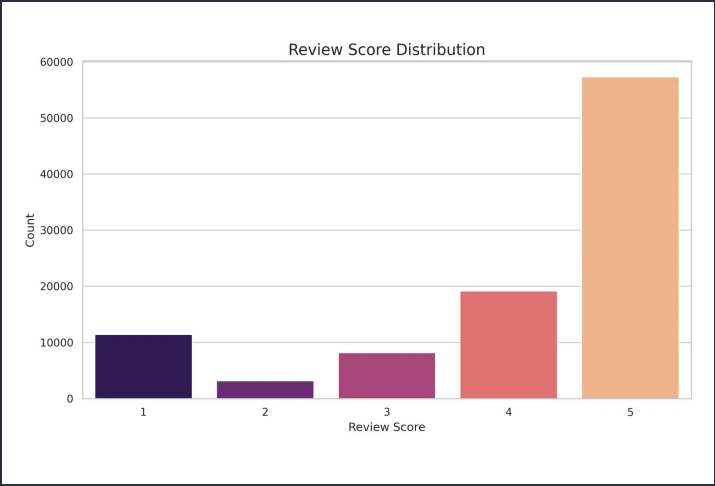
Monday records the highest order volume, suggesting a weekend-to-weekday shopping transition.

PURCHASE HOUR



Customer activity peaks during afternoon and evening hours (10:00 - 22:00), indicating prime marketing windows.

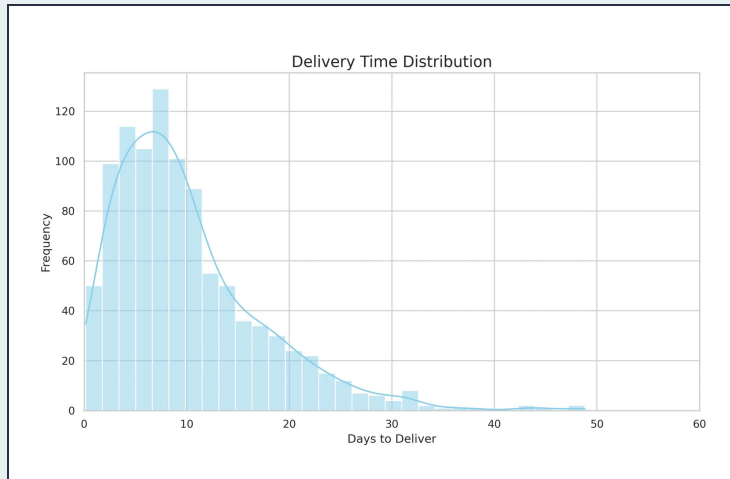
REVIEW DISTRIBUTION



Most customers give 5-star ratings, indicating high overall satisfaction across the platform.

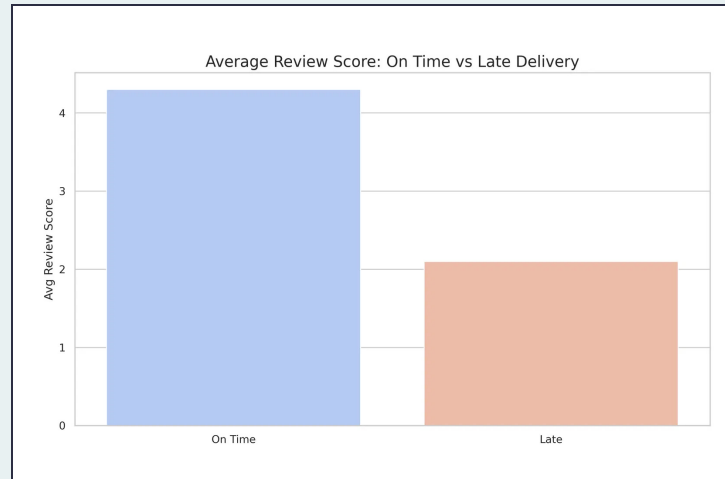
LOGISTICS EXCELLENCE IS THE KEY TO CUSTOMER SATISFACTION

DELIVERY TIME



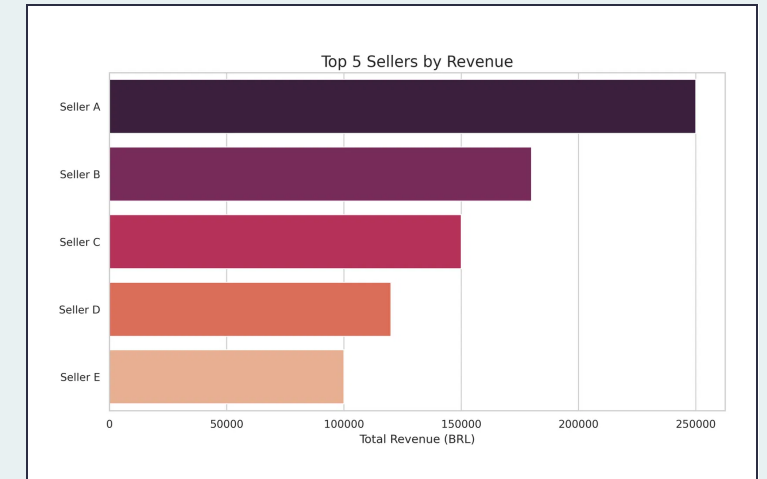
Most orders reach customers within 20 days, with an average delivery window of 12 days.

LATE DELIVERY IMPACT



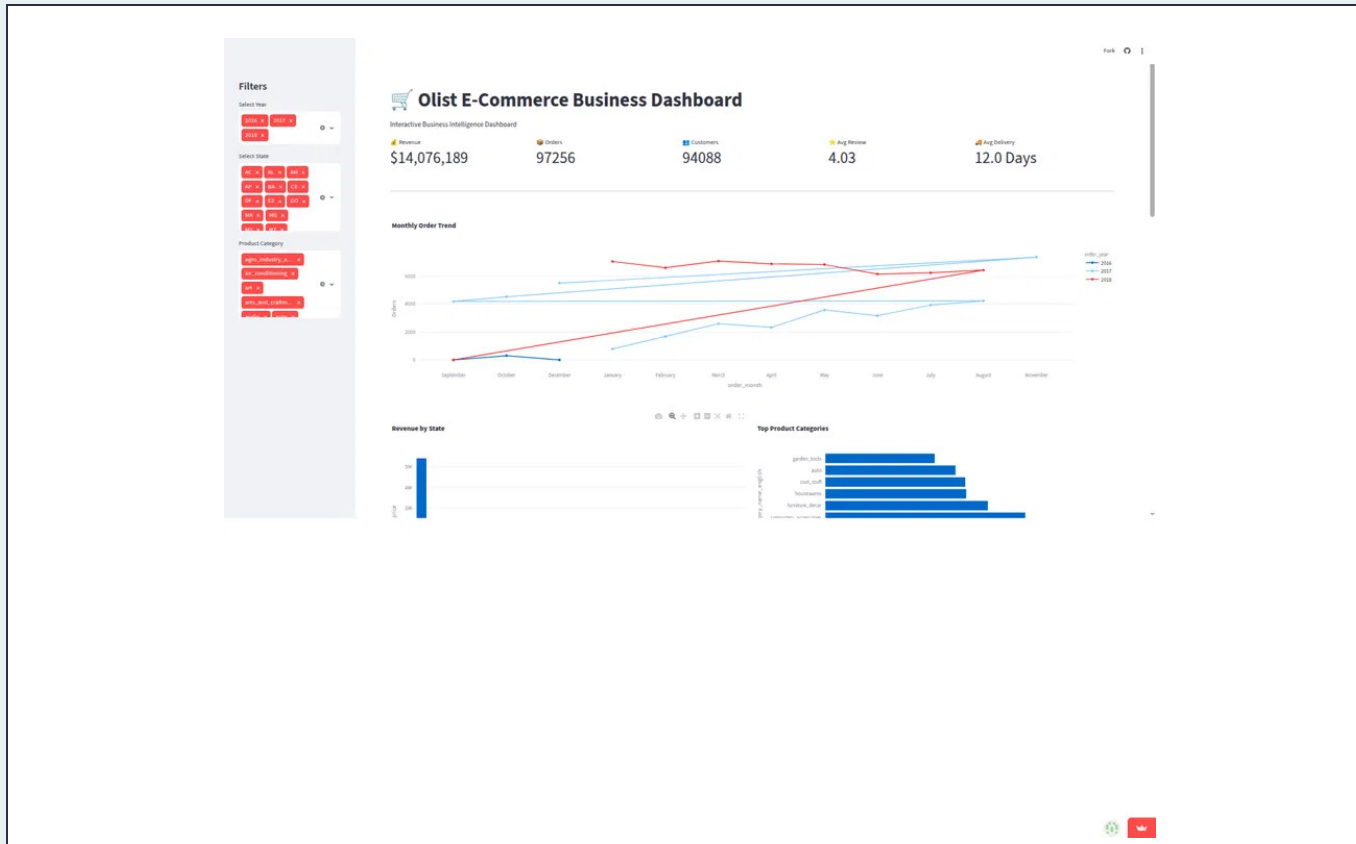
Late deliveries cause a drastic drop in review scores (from 4.3 to 2.1), highlighting a critical business risk.

REVENUE CONCENTRATION



A small cohort of top sellers drives a significant portion of total marketplace revenue.

STREAMLIT DASHBOARD: EMPOWERING REAL-TIME DECISION MAKING



KEY FEATURES

-  Interactive Global Filters
-  Real-time KPI Tracking
-  Dynamic Plotly Visualizations
-  Seller & Category Drill-downs
-  Revenue & Review Analysis

The dashboard provides a centralized interface for real-time interactive exploration of critical business metrics, bridging the gap between raw data and strategic action.

STRATEGIC CONCLUSIONS: OPTIMIZING E-COMMERCE OPERATIONS



LOGISTICS EXCELLENCE

Delivery speed is the single most critical factor for maintaining high customer satisfaction. Reducing late deliveries is vital for protecting brand reputation.



DEMAND OPTIMIZATION

Operational resources and marketing spend should be optimized for Monday peaks and afternoon traffic surges to maximize conversion rates.



GROWTH STRATEGY

Revenue growth is highly concentrated. Identifying and providing targeted support to top-performing sellers is essential for long-term platform success.



TECHNOLOGY INTEGRATION

Interactive dashboards bridge the gap between complex datasets and business action, enabling stakeholders to make faster, data-driven decisions.

THANK YOU

ANY QUESTIONS?

Somendar Kumar Das

Data Visualization – Summer 2026